

Chapter 7. Other Requirements

“Fast, Cheap, Good : Choose any two”

↳ 세 개 다 만족시킬 수 없다.

Objectives

- Show Supplementary Specification, Glossary, Vision & Business Rules.
- Compare and contrast system features with use cases.
- Define quality attributes.

(Use Case 이외의 요구사항 artifact)

Other Requirements Artifacts (Use Case 이외의)

USE CASE 이외의 다른 요구사항 (quality requirement) 요구사항 산출물

- **Supplementary Specification**

- Captures and identifies other kinds of requirements, such as
 - URPS+ (usability, reliability, performance, supportability, +)
 - constraints, standards, licensing, and so forth.

↑
제약들

- **Glossary** 용어정의, 데이터 사전

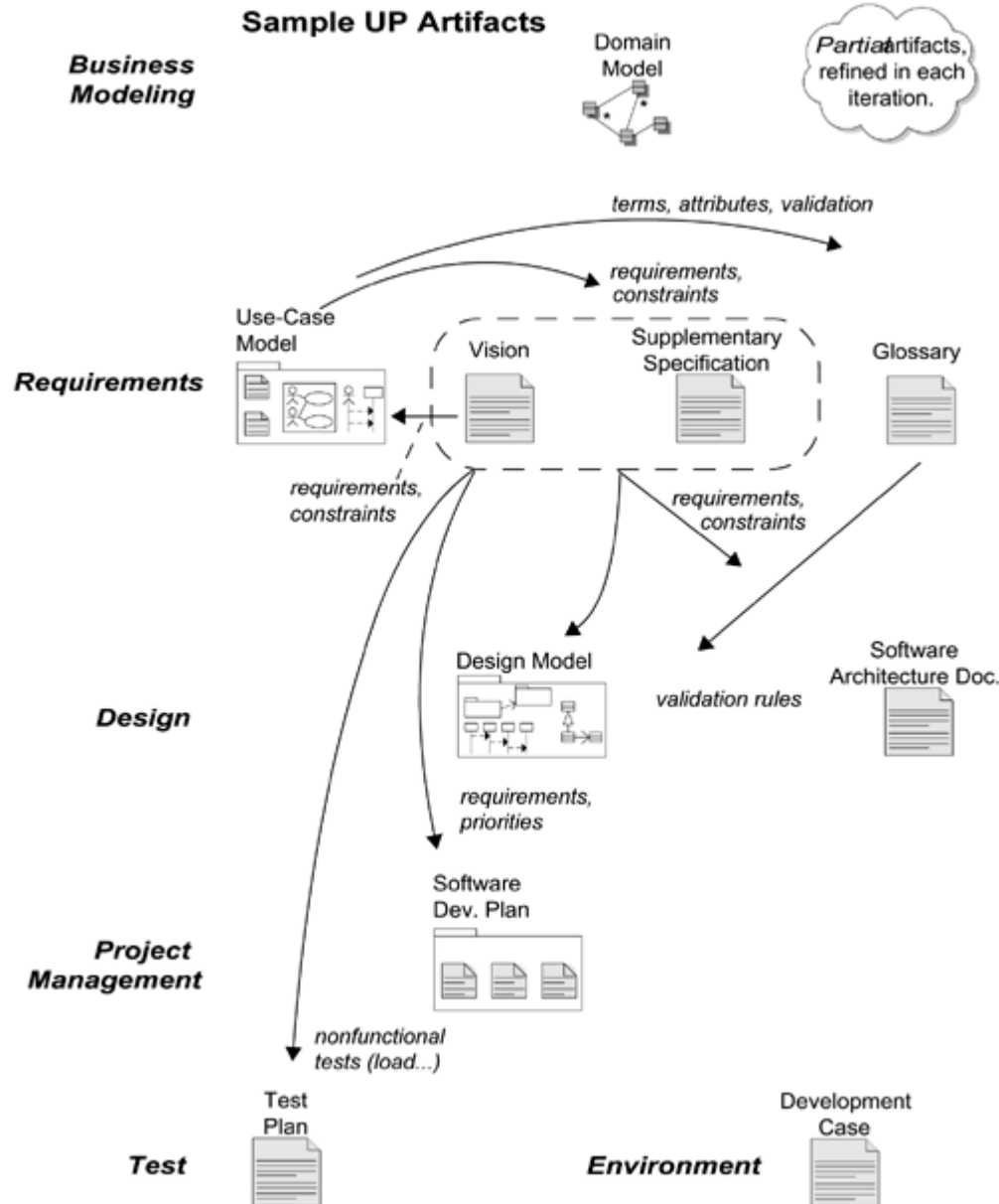
- Captures terms and definitions; a data dictionary

- **Vision** 프로젝트 비전 요약

- Summarizes the “vision” of the project; an executive summary

- **Business Rules** 도메인 규칙/정책

- Capture long-living and spanning rules or policies (such as tax laws), that transcend one particular application



Supplementary Specification

→ URPS+,
→ quality requirement

- It captures other requirements, information and constraints not easily captured in the use cases or Glossary, including system-wide “URPS+” quality attributes or requirements.

+ 품질 관련 속성,

+ 일반적인 기능.

- Elements of the Supplementary Specification include:

- Common functionality, usability, reliability, performance, and supportability
- reports
- hardware and software constraints (operating and networking systems, ...)
- development constraints (for example, process or development tools)
- other design and implementation constraints
- internationalization concerns (units, languages)
- documentation (user, installation, administration) and help
- licensing and other legal concerns
- packaging
- standards (technical, safety, quality)
- physical environment concerns (for example, heat or vibration)
- ...

→ 3기, 예러처리, 보안, 로그인, 예러처리, 보안.

Analyze These Thoroughly During Inception Only?

- No. The UP is an iterative and evolutionary method.
 - Means that production-quality programming and testing should happen very early, long before most requirements have been fully analyzed or recorded.
 - Feedback from early programming and tests evolve the requirements.
- However, research shows that
 - useful to have a high-level "top ten" list of coarse-grained requirements near the start.
 - also useful to spend non-trivial early time understanding the non-functional requirements (such as performance or reliability), as these have a significant impact on architectural choices.

Example: Supplementary Specification

Revision History

Version	Date	Description	Author
Inception draft	Jan 10, 2031	First draft. To be refined primarily during elaboration.	Craig Larman

Introduction

This document is the repository of all NextGen POS requirements not captured in the use cases.

Functionality (*common across many use cases*)

Logging and Error Handling

Log all errors to persistent storage.

↳ 밑바탕적인 기능.

Pluggable Rules

At various scenario points of several use cases (to be defined) support the ability to customize the functionality of the system with a set of arbitrary rules that execute at that point or event.

Security

All usage requires user authentication.

Usability

Human Factors

The customer will be able to see a large-monitor display of the POS. Therefore:

- Text should be easily visible from 1 meter.
- Avoid colors associated with common forms of color blindness.
 - Speed, ease, and error-free processing are paramount in sales processing, as the buyer wishes to leave quickly, or they perceive the purchasing experience (and seller) as less positive.
 - The cashier is often looking at the customer or items, not the computer display. Therefore, signals and warnings should be conveyed with sound rather than only via graphics.

NextGen POS Example

Reliability

Recoverability

If there is failure to use external services (payment authorizer, accounting system, ...) try to solve with a local solution (e.g., store and forward) in order to still complete a sale. Much more analysis is needed here...

Performance

As mentioned under human factors, buyers want to complete sales processing very quickly. One bottleneck is external payment authorization. Our goal: authorization in less than 1 minute, 90% of the time.

Supportability

Adaptability

Different customers of the NextGen POS have unique business rule and processing needs while processing a sale. Therefore, at several defined points in the scenario (for example, when a new sale is initiated, when a new line item is added) pluggable business rule will be enabled.

Configurability

Different customers desire varying network configurations for their POS systems, such as thick versus thin clients, two-tier versus N-tier physical layers, and so forth.

....

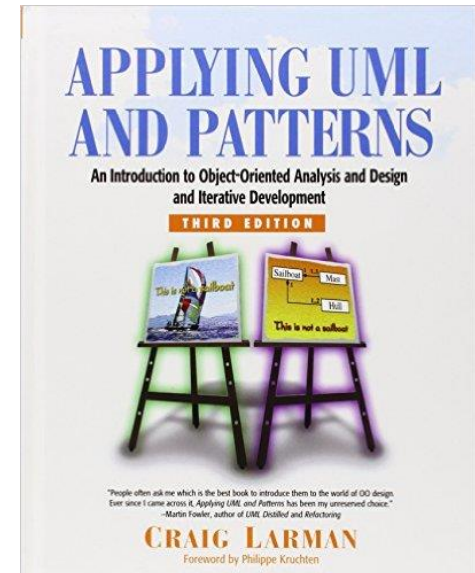
Implementation Constraints

NextGen leadership insists on a Java technologies solution, predicting this will improve long-term porting and supportability, in addition to ease of development.

...

NextGen POS Example

- Supplementary Specification
- Vision
- Glossary
- Business Rules (Domain Rules)
- Refer to the book for concrete examples



Process: Evolutionary Requirements in Iterative Methods

Discipline	Artifact	Inception	Elaboration	Construction	Transition
Business Modeling	Domain Model		s		
Requirements <i>inception은 시작, elaboration은 세부화</i>	Use-Case Model	s	r		
	Vision	s	r		
	Supplementary Specification	s	r		
	Glossary	s	r		
Design	Design Model		s	r	
	SW Architecture Document		s		